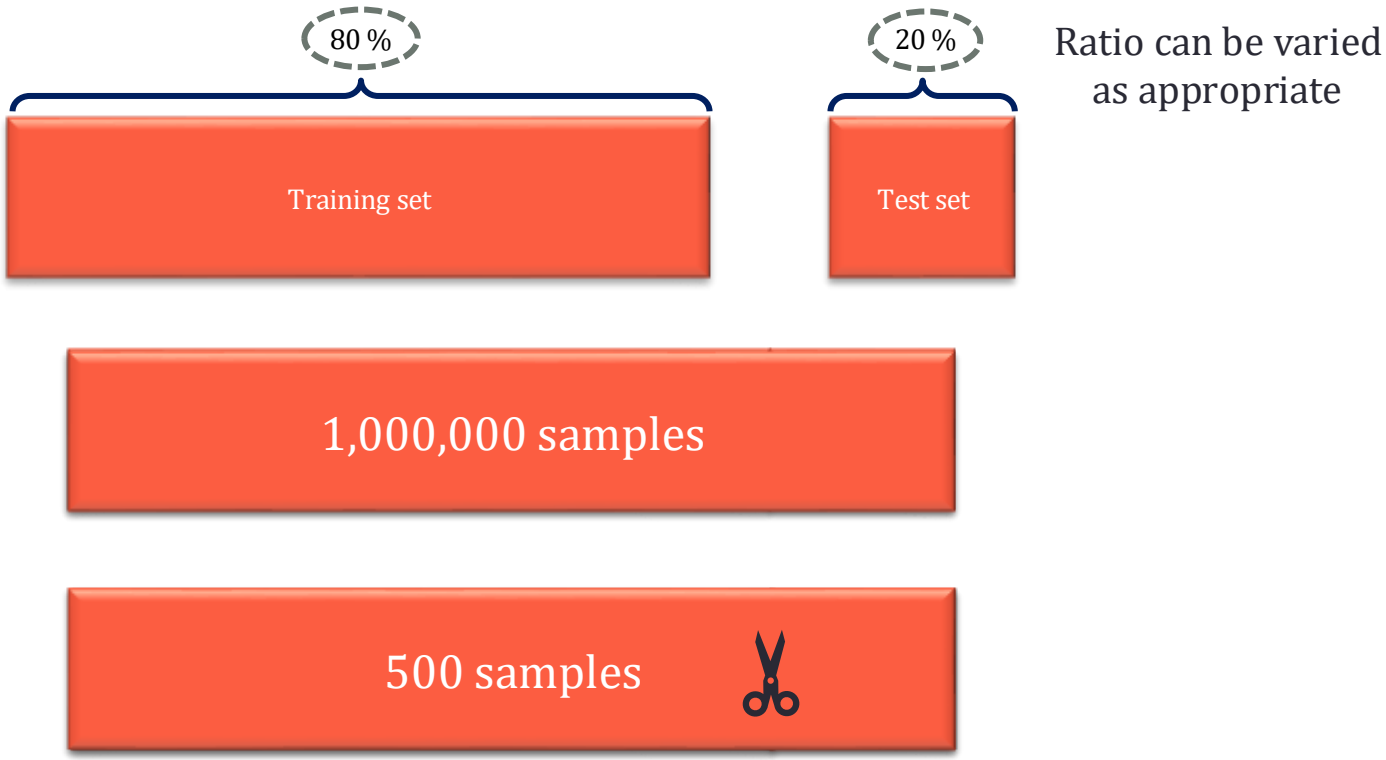
Loose ends from HW2

- Hyper-parameters
  - something you decide
  - bin size = ? Parametric or non-parametric? number of parameter
  - Tuned on validation set
- Parameters
  - something that is learned by the algorithm
  - weights in linear regression, probability in histogram bins, Gaussian means
- A majority class baseline
  - Powerful if one class dominates. Often happens in real life
  - Recognizer becomes biased towards the majority class (the prior term)
  - How to deal with this?

Splitting data

Simple train-test split



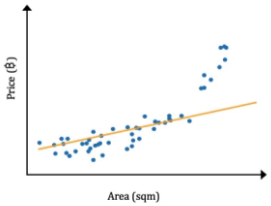
Stratified splitting – tries to keep the distribution in the training and test the same

Leave-one-out

Multiple train-test split

Estimate the true performance (expected performance)

Random

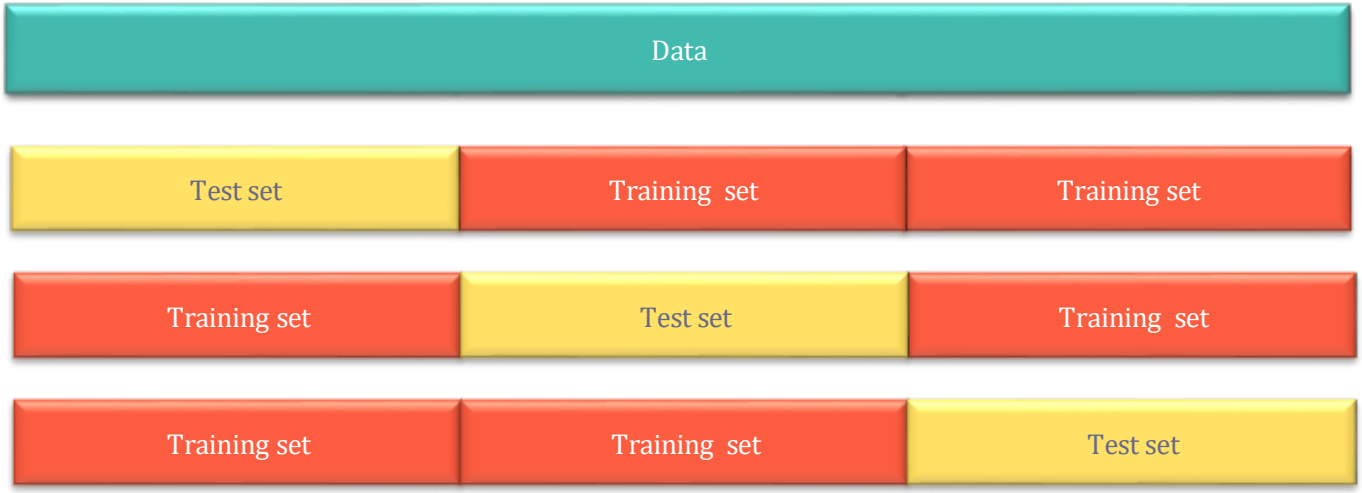


$\hat{y} = 2.3x1 - 3.4x2 + 4.2$

Train-test split → Train model → Predict

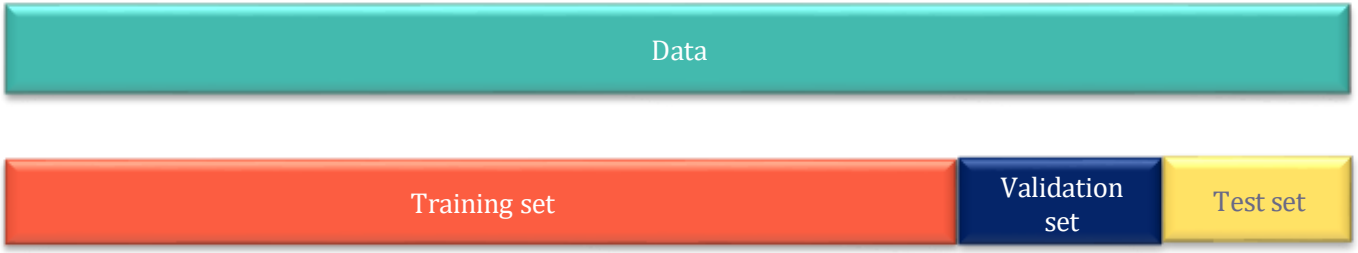
x 10

Cross-validation (CV)



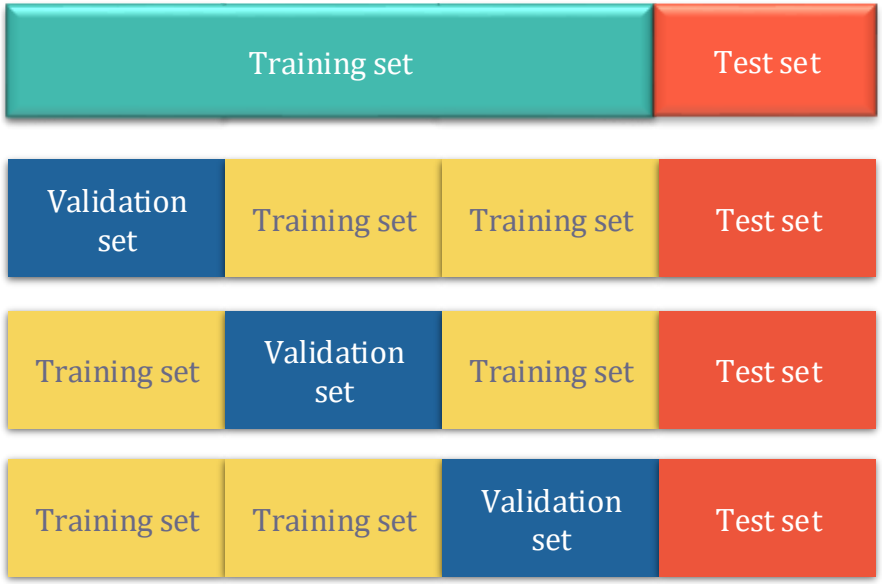
3-fold cross-validation

Estimating hyper-parameters

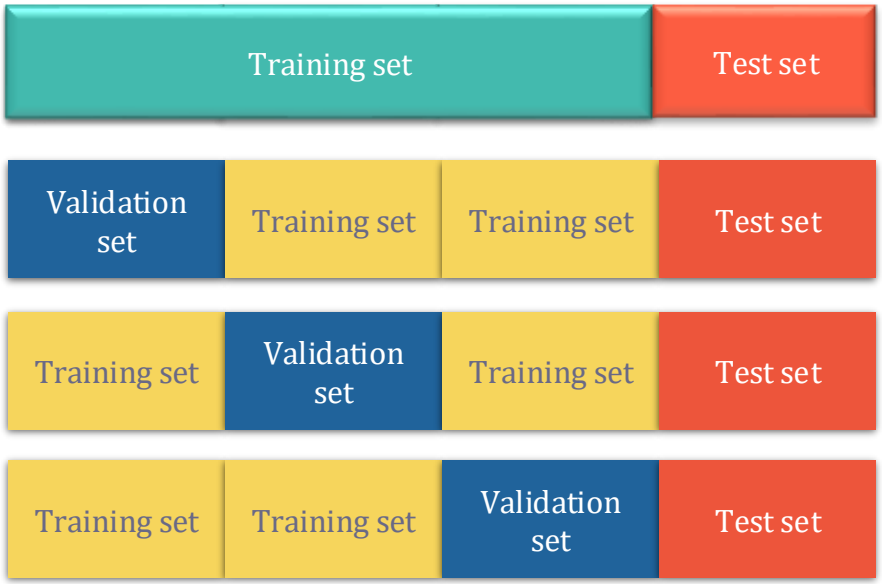


You decide on the models and hyperparameter on the validation performance
Don't cheat!

CV with validation set



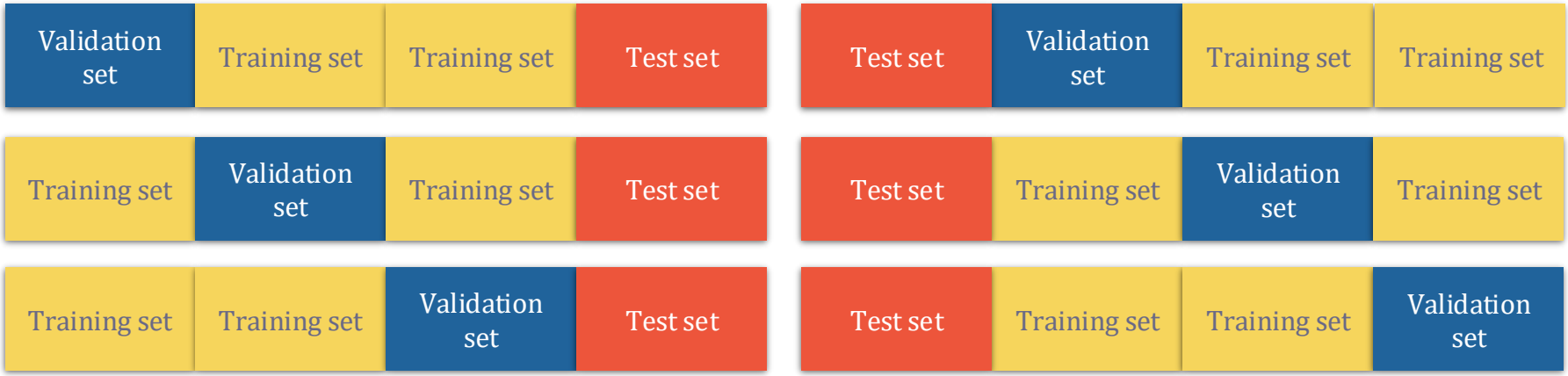
Estimating true performance of our pipeline



Freeze the test set
Touch the test set as less as possible

Nested CV

It the fixed test set is too small to give a reliable estimate, use nested CV.



Test split 1

Test split 2

In practice what's a good data size?

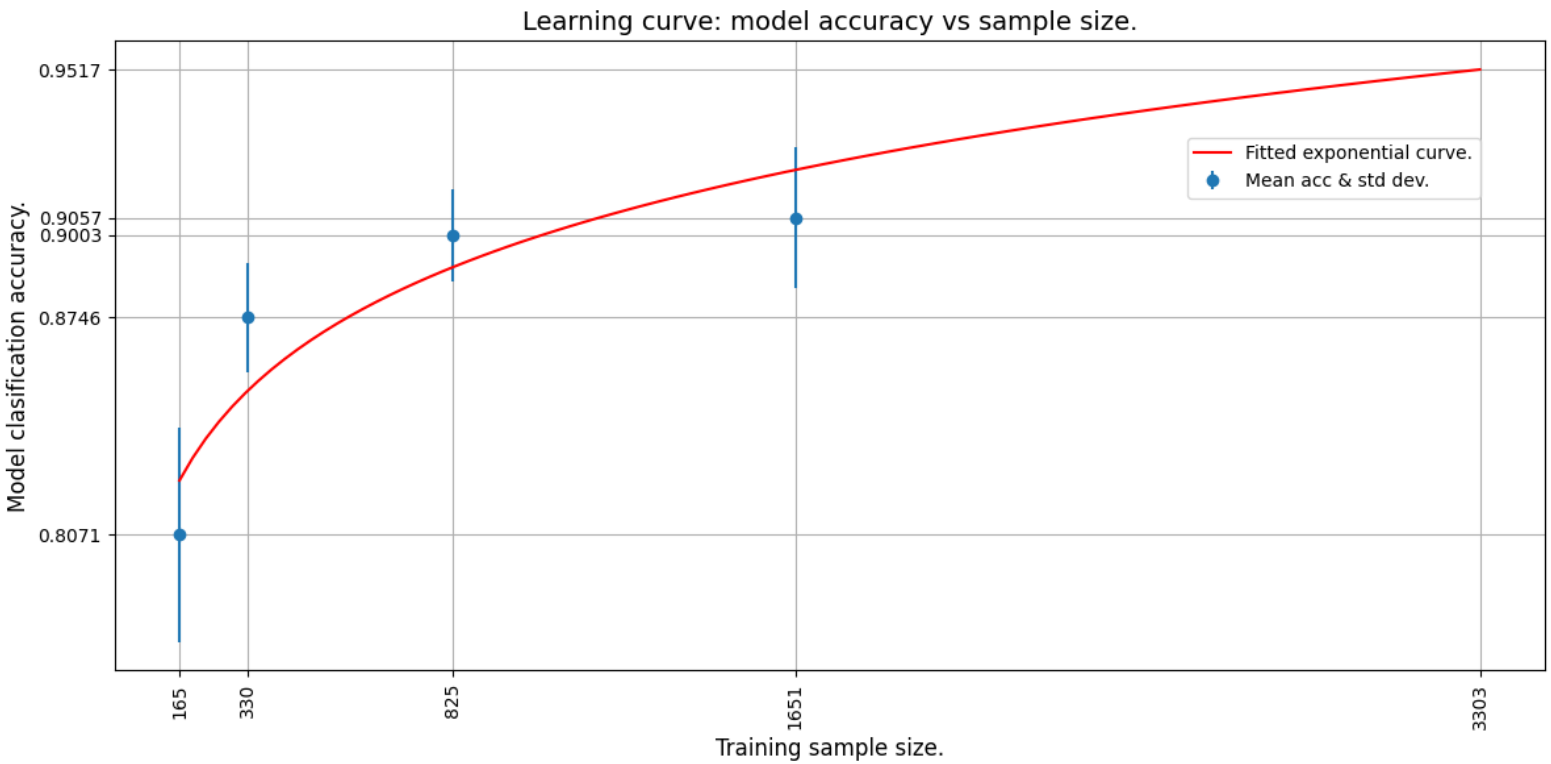
- These splitting are ideal for academic purposes.
- In real life, we usually have to collect data ourselves
- Guide on data collection
  - Collect test and validation data first. Make sure these data are close to the real use cases.
  - Training data can be slightly different from real use case if required.

In practice what's a good data size?

- Test data: estimate models' performance. Size should be enough to ensure a good enough estimate. Covers enough types of the expected use cases. Sample size estimate (from statistics) can also be helpful
- Validation data: select models' hyperparameter. Size should be enough to differentiate between different models. For example, if you care about a 0.1% difference in model performance, your validation data should be at least 1000 samples.
- Training data: train you model. Size should be enough to get you target performance. This is hard to estimate in advance. Literature review might help you. Otherwise, you might want to see how data size effect performance in your application.

Extra reading on sample size determination

- https://keras.io/examples/keras_recipes/sample_size_estimate/
- https://aclanthology.org/2023.findings-acl.419.pdf
- https://proceedings.iclr.cc/paper_files/paper/2025/file/a0709efe5139939ab69902884ecad9c1-Paper-Conference.pdf



RoC

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction
100	3	3	**Yes**	**0.8**
20	1	1	**Yes**	**0.3**
80	3	2	**No**	**0.6**
40	1	1	**No**	**0.2**
40	1	1	**No**	**0.1**

What happens if I set my threshold at 0.5?

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction	Metric
100	3	3	**Yes**	**0.8**	**TP**
20	1	1	**Yes**	**0.3**	**FN**
80	3	2	**No**	**0.6**	**FA**
40	1	1	**No**	**0.2**	**TN**
40	1	1	**No**	**0.1**	**TN**

What happens if I set my threshold at 0.5?

True positive rate =

False alarm rate =

Precision =

Recall =

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction	Metric
100	3	3	**Yes**	**0.8**	**TP**
20	1	1	**Yes**	**0.3**	**FN**
80	3	2	**No**	**0.6**	**FA**
40	1	1	**No**	**0.2**	**TN**
40	1	1	**No**	**0.1**	**TN**

What happens if I set my threshold at 0.5?

True positive rate = ½

False alarm rate = ⅓

Precision = ½

Recall = ½

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction	Metric
100	3	3	**Yes**	**0.8**	
20	1	1	**Yes**	**0.3**	
80	3	2	**No**	**0.6**	
40	1	1	**No**	**0.2**	
40	1	1	**No**	**0.1**	

What happens if I set my threshold at 0.15?

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction	Metric
100	3	3	**Yes**	**0.8**	**TP**
20	1	1	**Yes**	**0.3**	**TP**
80	3	2	**No**	**0.6**	**FA**
40	1	1	**No**	**0.2**	**FA**
40	1	1	**No**	**0.1**	**TN**

What happens if I set my threshold at 0.15?

True positive rate = 1

False alarm rate = 2/3

Precision = 2/4

Recall = 2/2

Prediction and thresholds

Beer	Grass	Rice	Flood	Prediction	Metric
100	3	3	**Yes**	**0.8**	
20	1	1	**Yes**	**0.3**	
80	3	2	**No**	**0.6**	
40	1	1	**No**	**0.2**	
40	1	1	**No**	**0.1**	

What happens if I set my threshold at 0.5?

True positive rate = ½

False alarm rate = ⅓

Precision = ½

Recall = ½

What happens if I set my threshold at 0.15?

True positive rate = 1

False alarm rate = 2/3

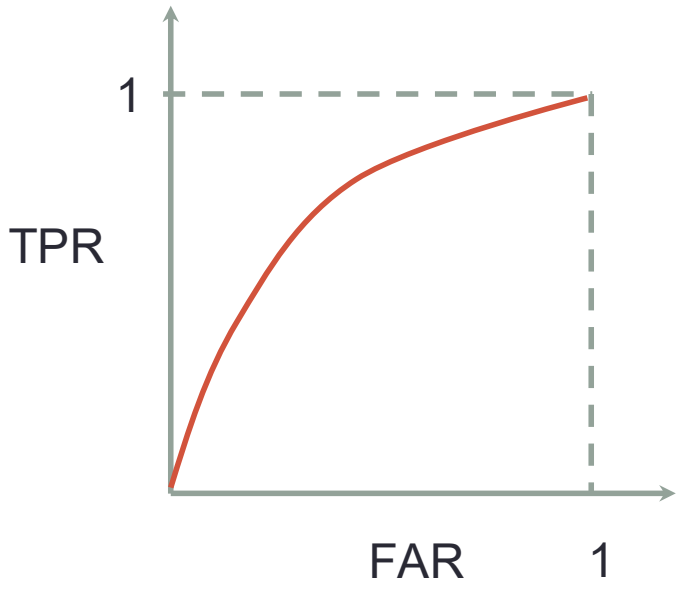
Precision = 2/4

Recall = 2/2

Receiver operating Characteristic (RoC) curve

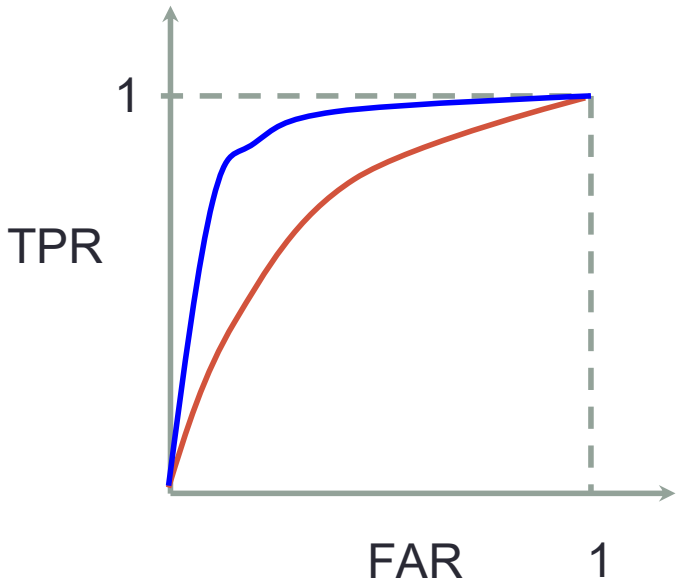
- Describes what happen when we change the threshold
- FA<->TP is a tradeoff

  This is why we need to think of the application when thinking of metrics.
- Plot FA rate and TP rate as threshold changes



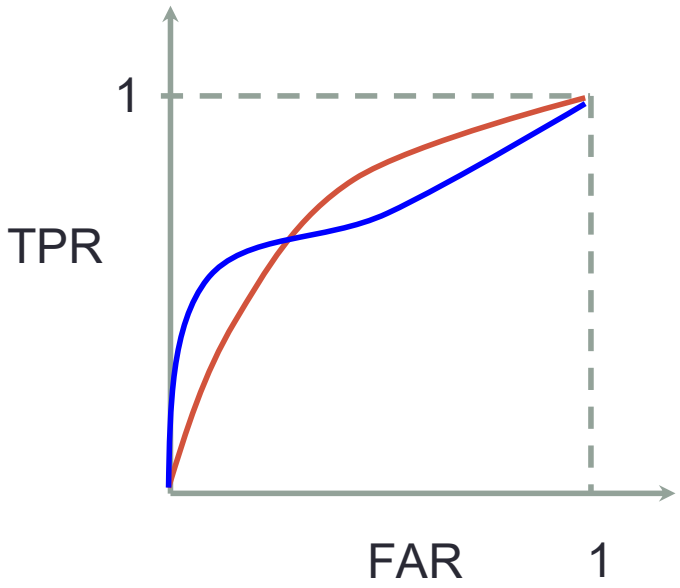
Comparing detectors

- Which is better?



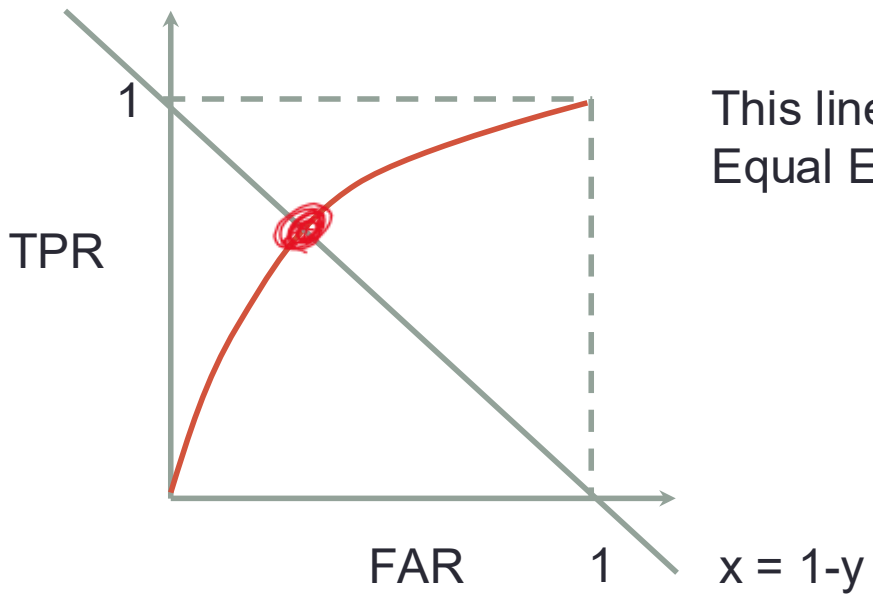
Comparing detectors

- Which is better?



Selecting the threshold

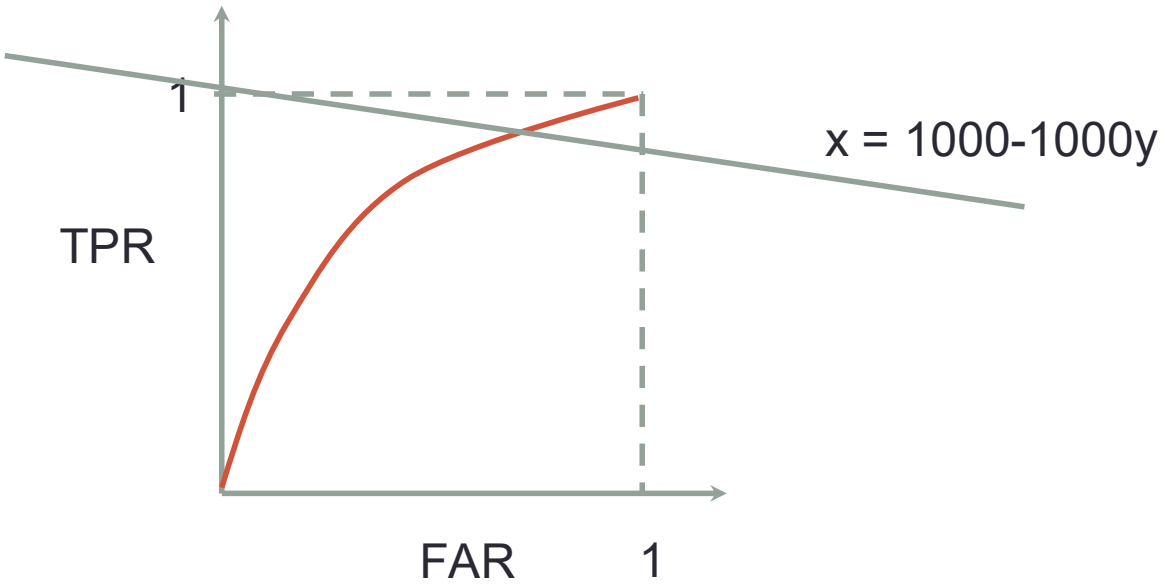
- Select based on the application
- Trade off between TP and FA. Know your application, know your users.
  - A miss is as bad as a false alarm FAR = 1-TPR => x = 1-y



This line has a special name
Equal Error Rate (EER)

Selecting the threshold

- Select based on the application
- Trade off between TP and FA. Know your application, know your users. Is the application about safety?
  - A miss is 1000 times more costly than false alarm.
    - FAR = 1000(1-TPR) => x = 1000-1000y



Churn prediction

Predict whether a customer will stop subscription, so we can send a promotional ad.

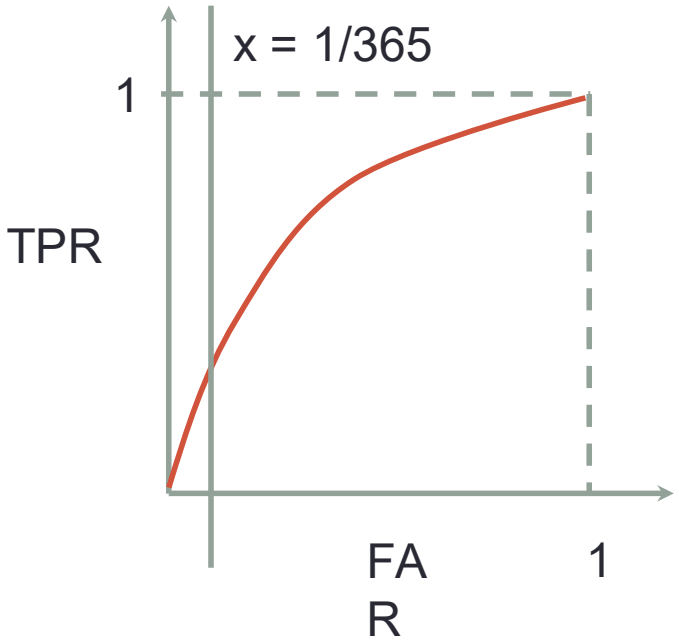
Usual subscription fee 50
Cost of calling the customer 5

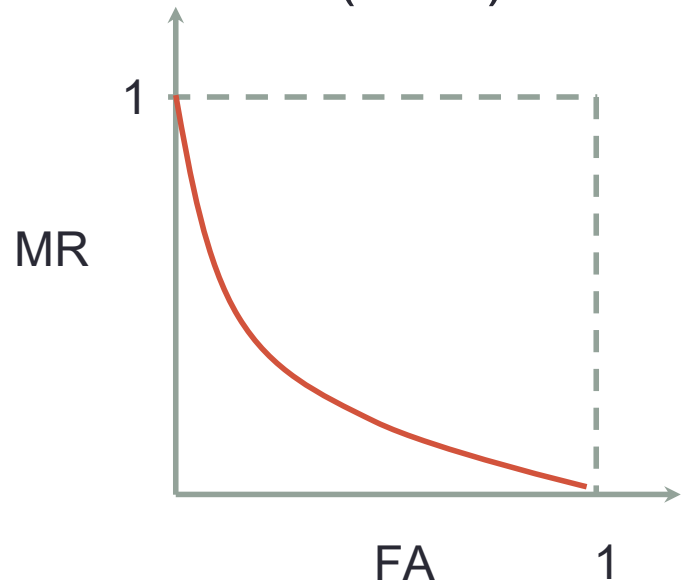
Promotional subscription fee 25

Describe the strategy to pick the threshold

Selecting the threshold

- Select based on the application
- Trade off between TP and FA.
  - Regulation or hard threshold
  - Cannot exceed 1 False alarm per year
    - If 1 decision is made everyday, FAR = 1/365



Notes about RoC

- Ways to compress RoC to just a number for easier comparison -- use with care!!
  - EER
  - Area under the curve
  - F score
- Other similar curve - Detection Error Tradeoff (DET) curve
- PR Curve
  - Plot Precision vs Recall
- Can plot on log scale for clarity